

Satellites Beam Hopping Scheduling for Interference Avoidance

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Abstract—The deployment of low earth orbit (LEO) satellites megaconstellations presents a promising way for achieving global coverage and service, attributed to their comparatively low round-trip latency and launch costs. However, this surge in LEO satellite launches exacerbates the scarcity of the limited spectrum resources. Spectrum sharing between satellite constellations and terrestrial networks and beam hopping (BH) technology emerge as viable strategies to mitigate this spectrum shortage. To enhance spectrum efficiency and avoid serious inter-system interference, we investigate the beam hopping scheduling of satellites for interference avoidance. The beam hopping scheduling of the integrated satellite-terrestrial wireless networks system is formulated as throughput-driven beam hopping (TDBH) problem and satisfaction-rate-driven beam hopping (SDBH) problem, respectively. In particular, we decompose the TDBH problem into two sub-problems by relaxation, and a genetic algorithm (GA) is introduced to handle the SDBH problem. The impact of channel conditions and traffic load intensity on the satellite system throughput is analyzed in TDBH simulation. As for SDBH optimization problem, the simulation results show that the proposed GA algorithm improves the average traffic satisfaction rate by 16.96% at least, compared with other benchmarks and suits to scenarios with different traffic demands and fading channel conditions.

Index Terms—Multi-beam satellite, beam hopping scheduling, interference avoidance.

I. INTRODUCTION

THE satellite communication system is an indispensable complement to next-generation wireless communications networks, facilitating communication services in remote and challenging-to-reach areas [1]. A single-beam satellite covering the ground with a wide beam is no longer sufficient to meet the increasing demands for data rates and connection density from ground users. Compared with single-beam satellites, the narrow beams of multi-beam satellites concentrate transmitting

power and improve the signal-to-noise ratio, and the system throughput can be further improved. However, conventional multi-beam satellites are equipped with fixed beams and transponders. For this reason, the satellite cannot dynamically allocate resources. Under this circumstance, it can be a waste of resources due to the non-uniformity of traffic requests from users which may change from slot to slot. In addition, the very limited payload resources of satellites prevent the simultaneously narrow-beam services to the cells within the satellite-coverage area.

A promising solution to this problem is the implementation of beam hopping (BH) technology, which enables satellites to serve multiple cells with a limited number of beams in a time-division manner [2], [3]. With multi-port amplifiers and phased array antennas, the beam directions, sizes, and shapes of satellites can be controlled. As a result, satellites can flexibly allocate bandwidth and power resources for beams. Such flexibility is crucial to allocate bandwidth, power, and other communication resources to fulfill the quality-of-service requirements of cells while efficiently utilizing the limited communication and computational storage resources aboard the satellite. However, given the vast coverage of satellites, requests from different cells can be time-varying and exhibit heterogeneity, adding complexity to the resource allocation problem. Therefore, the key issue lies in how to design the BH pattern, which determines the activated beams in each time slot, to match the limited resources to the demands of each cell [2], [4].

In order to support BH in multi-beam satellites, some super-framing specifications have been proposed in DVB-S2X standard [5]. Furthermore, to achieve compatibility, the International Telecommunication Union (ITU) conducted a detailed study on the resource management as well as control of satellite and ground systems [6]. Although ITU formulated frequency use rules, with an increasing number of data services and massive number of low earth orbit (LEO) satellites [7], radio resources in S-band (within FR1 in terrestrial network) and Ka-band (within FR2 in terrestrial network) have been exhausted for both non-terrestrial network and terrestrial networks [8], which makes frequency resources even more scarce. Under such circumstance, satellite networks will inevitably share the spectrum with terrestrial networks.

The geosynchronous Earth orbit (GEO) system remains relatively stationary with respect to the Earth. However, when an active LEO satellite moves into the coverage area of a GEO satellite, co-interference becomes a problem for these two independent satellite systems. According to [6] and [9], the operation of a low orbit satellite cannot affect the operation

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of a high orbit satellite. Besides interference between satellites, unacceptable interference between satellites and ground users may also occur when spectrum resources are shared by satellite systems and terrestrial networks, leading to the uncertainty of quality of service (QoS).

As non-renewable resources, frequency bands will seriously restrict the development of future aerospace-integrated networks. Tremendous efforts have been devoted to overcoming the scarcity of spectrum resources and improving spectrum efficiency. Some researches focus on power allocation optimization to avoid frequency interference [10], [11], [12]. Reference [13] proposes eight spectrum sharing schemes for GEO satellite networks to produce the frequency hopping sequence in the earth-moving LEO network. Reference [14] improves the spectrum efficiency of mmWave satellite-terrestrial communication system through the optimization of the beamwidth. An intelligent spectrum sharing scheme is presented to improve spectrum utilization rate of satellite-terrestrial spectrum-shared networks in [15]. Ding et al. [16] proposes a beam-management method in which interference beams of LEO satellites can be turned off and switched to decrease the interference on the GEO user.

The shortage of frequency resources leads to interference of the satellite system. To address this problem in BH scheduling, beam patterns with spatial isolation can be selected to avoid the inter-beam interference [17]. Reference [2] designs BH patterns with spatial isolation to avoid the interference of both inter-beam and inter-satellite. Spatial isolation is also used to avoid inter-cluster interference [18]. Based on the spatial isolation, non-orthogonal multiple access (NOMA) is used to reduce the interference for cell edge users [19]. In addition, many scheduling methods based on beam hopping have been proposed under their satellite system models [20], [21], [22]. Meng et al. [23] proposes a beam hopping allocation algorithm in downlink multicarrier direct-sequence code division multiple access (MC-DS-CDMA) scenario to minimize the energy consumption without consideration of non-uniformly distributed users. Reference [24] investigates a theoretical model of beam hopping in the uplink transmission based on discrete-time queueing theory to derive and analyze the explicit expressions of various user performance metrics. The minimum beam illumination probability of specific performance objectives can be obtained to provide guidance for system design. Reference [25] focuses on the beam hopping scheduling with hybrid precoding to optimize the transmit power with the requirement of rate demands. To reduce the complexity of the BH scheduling methods and enhance the flexibility of resource allocation, deep reinforcement learning has been employed properly, as an unsupervised learning method to solve the sequential decision-making problems [26], [27], [28]. However, these beam hopping scheduling strategies vary from the system models and optimization problems.

Different from the above BH optimization problems, this paper proposes two BH scheduling problems for interference avoidance, according to whether the traffic demands can be adjusted. In addition, the interference is characterized by an interference pattern in these optimization problems to comprise the co-interference between satellites and co-interference

between satellites and terrestrial networks. Extensive simulation results evaluate the performance of the proposed methods. The main contributions and innovations of this work are summarized as follows:

- In an integrated satellite-terrestrial wireless network system, this paper aims to design the BH patterns of satellites to maximize the system throughput and the satisfaction rate of traffic demand with the requirement of interference avoidance, considering the downlink transmission of satellites. First, we formulate two BH optimization problems for the above two objectives according to whether the traffic demands can be adjusted, namely, throughput-driven beam hopping (TDBH) and satisfaction-rate-driven beam hopping (SDBH). The interference constraints in these two optimization problems are, on one hand, assumed to be obtained in advance by the controller of the earth station. On the other hand, they are considered as an interference pattern that specifies which cells cannot be served by a satellite in each time slot. We consider that the satellite which employs TDBH provides a service to the ground cells whose traffic demands can be regulated, therefore the adaptive rate adjustment factor is defined in the TDBH scheduling problem. In contrast, the user's traffic demands under the SDBH satellite service cannot be regulated. In addition, to guarantee the fairness of the service, the minimum value of the adaptive rate adjustment factor is constrained in the TDBH optimization problem. Simultaneously, the unserved time of target cells is defined in the SDBH optimization problem.
- The original TDBH optimization problem is decomposed into two sub-problems through relaxation, which aims to reduce the complexity. Simulation analyzes the impact of three Shadowed-Rician channel conditions and traffic load intensity on the performance of the satellite throughput.
- The proposed genetic algorithm (GA) method is devised to deal with the SDBH optimization problem in this paper. The impacts of Shadowed-Rician fading channel and uniform traffic distributions are taken into consideration in the simulation.

The rest of the paper is organized as follows. Section II describes the integrated satellite-terrestrial wireless networks system model. Section III elucidates the relationship between TDBH and SDBH, then formulates the TDBH optimization problem and SDBH optimization problem with the requirement of interference avoidance. In Section IV, the solutions to TDBH and SDBH problems are presented. Simulations validate the effectiveness of the proposed methods in Section V. Section VI concludes this paper.

II. SYSTEM MODEL

A. Co-Existing Multi-Beam Satellites System

Assume that both LEO satellite and GEO satellite employ beam hopping technology in the integrated satellite-terrestrial wireless networks and they are controlled by their earth stations independently. Due to the scarcity of spectrum resources, the GEO system, the LEO system and terrestrial networks

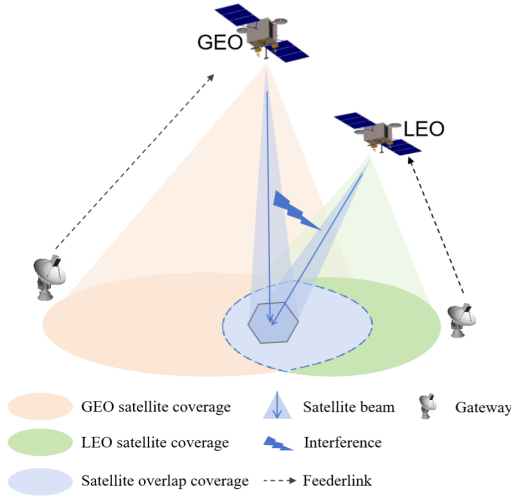


Fig. 1. Spectrum sharing in integrated satellite-terrestrial wireless networks system.

have to share the bandwidth inevitably. Under this situation, as illustrated in Fig. 1, when the LEO satellite moves into the coverage range of the GEO satellite, the LEO beams and GEO beams in the overlapping area may cause serious co-frequency interference, and also interfere with the communication of ground users. This paper focuses on satellites beam hopping scheduling with two different optimization goals for interference avoidance. It is assumed that satellites can obtain interference information for the next scheduling period via earth station, including interference between high-orbit satellites and low-orbit satellites, as well as interference information between satellites and terrestrial networks. Then they generate interference avoidance patterns for each time slot, that is, they determine which cells cannot be served in each time slot. For example, LEO satellites beam hopping scheduler may receive an interference avoidance pattern based on GEO satellites scheduling result, as non-geostationary-satellites should not cause unacceptable interference to geostationary-satellite networks. In this case, the GEO satellite may adopt a BH strategy to maximize its performance, e.g., throughput, while LEO satellites should take interference avoidance into account for its beam selection.

B. Signal Model

As depicted in Fig. 2, this work considers a communication downlink between a satellite (LEO or GEO) and terrestrial virtual cells within the coverage area. Assuming that there are N virtual cells within the coverage area of the satellite. The satellite serves these cells in a time-division manner based on instructions from the ground control center, with k cells being served in each time slot at most. According to the DVB-S2X standard, the duration of a time slot for LEO is 2 ms . A period of BH scheduling is assumed to contain N_s time slots. The LEO satellite has a very limited service duration for each satellite pass of a considered area. Therefore, the time-varying characteristics of cell traffic demand can be neglected during a scheduling period. The traffic demand of the n -th cell is denoted as d_n (bps), the capacity from beam k to the cell it

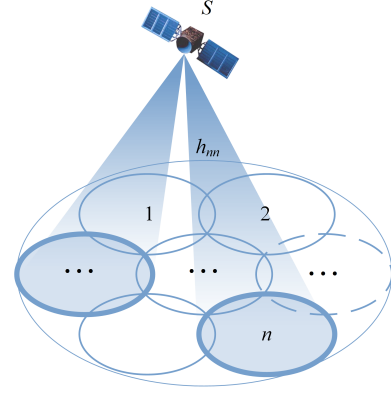


Fig. 2. Downlink transmissions of a multi-beam satellite communication system.

covers is assumed as the capacity from beam k to the center of the served cell. Let the transmitting signal at satellite S to N beam cells is $\mathbf{x}^{N \times 1}$, the received signal $\mathbf{y}^{N \times 1}$ can be expressed as [29],

$$\mathbf{y} = \sqrt{\mathbf{P}_r} \mathbf{H} \mathbf{x} + \mathbf{u}, \quad (1)$$

where $\mathbf{P}_r = \text{diag}(P_{r1}, \dots, P_{rn})$ denotes the received signal power, $\mathbf{H}$ is downlink channel matrix, and $\mathbf{u}$ is the system noise at the receiver, which follows a complex Gaussian distribution of zero mean.

Due to the long propagation distance of satellite channels, it is necessary to consider both large-scale and small-scale fading, while taking the impact of path loss into account. In general, the path loss of satellite channels includes free space propagation loss and rain attenuation. To facilitate the analysis, the sky is assumed to be clear so rain attenuation is not considered. As a result, the received signal power can be represented as

$$P_{rk} = G_t + G_r - L_f + P_{tk} \quad [\text{dB}], \quad (2)$$

where L_f is the free space propagation loss, G_r represents the receiving antenna gain, G_t denotes gain of satellite transmitting antenna, and P_{tk} is the transmitting power of the k -th beam, respectively.

Generally, satellite communication channels are modeled as Gaussian channels or weak Rician multipath channels in satellite communication, but there is a lack of consideration for shadow fading caused by multi-path and terrain conditions in practical applications. Therefore, the Shadowed-Rician channel model in [30] is considered here. The Shadow-Rician channel is an additive channel model, which is composed of a line-of-sight (LOS) component affected by shadow fading and a non-line-of-sight (NLOS) component not affected by shadow effects. The Shadow Rician received signal $S_r(t)$ can be expressed as

$$S_r(t) = Z(t) \exp(j\theta_0) + A(t) \exp(j\alpha(t)) \quad (3)$$

where $Z(t)$ is the amplitude of LOS component and it is Nakagami- m distributed. While $A(t)$ is the amplitude of NLOS component and it is Rayleigh distributed. They are independent stationary random processes. θ_0 is a deterministic phase vector and $\alpha(t)$ is the stationary random phase

vector and it follows a uniform distribution on the interval $[0, 2\pi)$.

Let

$$\tilde{\mathbf{h}} = \{\tilde{h}_{kl}\}^{N \times N} \quad (4)$$

be the Shadowed-Rician distribution matrix, the downlink channel $\mathbf{H} = \{h_{kl}\}^{N \times N}$ can be expressed as

$$\mathbf{H} = \tilde{\mathbf{h}} \circ \begin{bmatrix} \sqrt{\alpha(\theta_{1,1})} & \dots & \sqrt{\alpha(\theta_{1,N})} \\ \sqrt{\alpha(\theta_{2,1})} & \ddots & \vdots \\ \vdots & \ddots & \sqrt{\alpha(\theta_{N-1,N})} \\ \sqrt{\alpha(\theta_{N,1})} & \dots & \sqrt{\alpha(\theta_{N,N})} \end{bmatrix}, \quad (5)$$

where $h_{kl} = \tilde{h}_{kl} \cdot \sqrt{\alpha(\theta_{k,l})}$ denotes the interference of cell l to cell k , $\theta_{k,l}$ is the off-axis angle, $\circ$ is the Hadamard product. $\alpha(\theta_{k,l})$ represents the interference gain, which is given by

$$\alpha(\theta_{k,l}) = \begin{cases} 1, & \text{if } l = k \\ 4 \left| \frac{J_1(k_w a \sin \theta_{k,l})^2}{k_w a \sin \theta_{k,l}} \right|, & \text{if } l \neq k \end{cases}, \quad (6)$$

where $J_1(\cdot)$ refers to Bessel function of the first kind, $k_w = 2\pi f/c$ denotes the wavenumber, and a represents the radius of the antenna aperture.

In order to improve the system capacity and make full use of spectrum resources, the entire available bandwidth B of satellite is shared among the active beams. As a result, we obtain the signal to interference plus noise rate (SINR)

$$\text{SINR}_k = \frac{P_t \cdot \|h_{kk}\|^2 \cdot \alpha^2}{K_b T_{\text{noise}} B + \sum_{l \neq k} P_t \cdot \|h_{kl}\|^2 \cdot \alpha^2}, \quad (7)$$

where T_{noise} is equivalent to noise temperature, and K_b represents the Boltzmann constant. The channel capacity of cell n at the i -th time slot can be further calculated as:

$$c_{in} = B \log_2(1 + \text{SINR}_n). \quad (8)$$

III. BEAM HOPPING SCHEDULING OPTIMIZATION PROBLEM FORMULATION

In this section, we formulate two beam hopping scheduling optimization problems, TDBH scheduling problem and SDBH scheduling problem. On one hand, these two scheduling problems are based on two different optimization objectives. TDBH aims to maximize the system throughput and SDBH aims to meet the traffic demands as much as possible. On the other hand, these two scheduling problems can be independent if there is an independent scheduling relationship between satellites using TDBH and SDBH. These two scheduling problems can also be non-independent. For example, a GEO satellite employs TDBH, while a LEO satellite employs SDBH, as the scenario set in the simulation section. In this case, the SDBH based LEO satellite scheduling needs to be based on the scheduling result of the TDBH scheduling, since the GEO satellites are the primary users [6]. That means cells of a LEO satellite can not be served if they cause interference to GEO satellites, thus they will be included in the interference pattern of a LEO satellite.

A. Throughput-Driven Beam Hopping Scheduling

The ground cell service traffic demand can be larger than the channel capacity of the satellite-ground link which experiences shadowing effects in practical scenarios. In order to avoid transmission interruptions caused by this factor, we consider that satellites provide a common service to the ground cells whose traffic can be regulated. Therefore, the adaptive rate adjustment factor is introduced in TDBH as

$$\mathbf{A}^{1 \times N} = \{A_n\}^{1 \times N}, A_n \in (0, 1]. \quad (9)$$

The equivalent traffic demands $\mathbf{T}_e$ of ground cells after rate adjustment can be expressed as

$$\mathbf{T}_e = \mathbf{T} \circ \mathbf{A}, \quad (10)$$

where $\mathbf{T} = [T_1, T_2, \dots, T_N]$ is the original traffic demands. $A_n = 1$ means that cell n can be served without adjustment. $A_n < 1$ indicates the traffic of cell n needs to be adjusted to a certain extent due to poor channel conditions of the satellite-ground link or excessive traffic demand of ground users. Then we get the downlink equivalent channel capacity $\mathbf{C}_e$ as follows.

$$\mathbf{C}_e = \frac{1}{N_{\text{slot}}} \sum_{i=1}^{N_{\text{slot}}} \mathbf{g}_i \circ \mathbf{c}_i, \quad (11)$$

where $\mathbf{g}_i$ means the allocation result at the i -th slot and can be expressed as $\mathbf{g}_i = [g_{i1}, g_{i2}, \dots, g_{iN}]$. $g_{in} = 1$ indicates that the cell n is served at the i -th slot. $\mathbf{c}_i = [c_{i1}, c_{i2}, \dots, c_{iN}]$ is the channel capacity of cells at the i -th slot. N_{slot} means the total time slots in the TDBH scheduling period.

The equivalent channel capacity and traffic demands of cells are measured in Mbit/s. We assume that the TDBH based satellite system is modeled as a M/M/1 queue, then the channel capacity can be seen as the service rate and traffic demands can be seen as arrival rate. The average waiting time is obtained as

$$\mathbf{W}_t = \frac{1}{\frac{1}{N_{\text{slot}}} \sum_{i=1}^{N_{\text{slot}}} \mathbf{g}_i \circ \mathbf{c}_i - \mathbf{T} \circ \mathbf{A}}, \quad (12)$$

we consider there is a maximum value of the average waiting time, by substituting (10) and (11) into (12), the expression for the average waiting time is given as:

$$0 \leq \mathbf{W}_t = \frac{1}{\mathbf{C}_e - \mathbf{T}_e} \leq \tau. \quad (13)$$

The equivalent traffic demands will be satisfied easily when traffic demands are small. When traffic demands are significantly less than the channel capacities, the full utilization of all active beams becomes unnecessary to meet traffic demands. In this case, the maximum will be easily achieved if the optimization goal is to merely maximize the equivalent traffic demands. We consider delay as an implicit optimization criterion for the optimization problem. Specifically, the optimization goal of the proposed TDBH problem is to maximize the inner product of the equivalent demands and channel capacity. It maximizes the equivalent demands and channel capacity simultaneously, thereby implicitly reducing

the delay. The considered optimization problem is formulated below:

$$\begin{aligned}
 P0: \quad & \max_{\mathbf{G}, \mathbf{A}} \mathbf{C}_e \cdot \mathbf{T}_e \\
 \text{s.t.} \quad & C_1: g_{in} \in \{0, 1\}, i = 1, 2, \dots, N_{slot}, \\
 & n = 1, 2, \dots, N, \\
 & C_2: \sum_{n=1}^N g_{in} \leq K, i = 1, 2, \dots, N_{slot}, \\
 & C_3: 0 \leq W_t \leq \tau, \\
 & C_4: \delta \leq A_n \leq 1, \delta \in (0, 1), \\
 & C_5: \phi_{in} \in \{0, 1\}, i = 1, 2, \dots, N_{slot}, \\
 & n = 1, 2, \dots, N.
 \end{aligned} \tag{14}$$

The objective function of the beam scheduling optimization problem $P0$ is to maximize the inner product of the equivalent channel capacity $\mathbf{C}_e$ and the equivalent traffic demands $\mathbf{T}_e$. $\mathbf{G} = \{g_{in}\}_{N_{slot} \times N}$ is the optimization variable, which is the schedule result of TDBH at time slot i . As it is shown in C_1 , g_{in} is a boolean variable and $g_{in} = 1$ indicates that the cell n is illuminated by the beam. Constraint C_2 restricts the maximum number of beams that the TDBH based satellite can generate every time slot. Constraint C_3 restricts the minimum average waiting time. To ensure the service fairness, the adaptive rate adjustment factor A_n should be larger than the minimum value δ , which is shown in C_4 . In C_5 , we define the interference pattern. $\phi_{in} = 0$ means cell n can not be served in time slot i due to the interference, while $\phi_{in} = 1$ means cell n is free to be scheduled.

B. Satisfaction-Rate-Driven Beam Hopping Scheduling

Different from the TDBH scheduling, it is assumed that satellites focus on serving cells whose traffic demands can not be altered in SDBH scenario. The optimization goal of the SDBH is to dynamically select cells in a scheduling period so that communication demands can be met as much as possible. The traffic demand of the n -th cell is denoted as d_n and a period of SDBH scheduling is assumed to contain N_s time slots. Therefore, the total demand is $D_n = d_n \times N_s$ during the whole scheduling period. The considered optimization problem is formulated below:

$$\begin{aligned}
 P1: \quad & \max_{\mathbf{Z}} \sum_{n=1}^N \frac{R_n}{D_n} \\
 \text{s.t.} \quad & C_1: z_{in} \in \{0, 1\}, \\
 & C_2: \sum_{n=1}^N z_{in} \leq K, i = 1, 2, \dots, N_s, \\
 & C_3: R_n = \sum_{i=1}^{N_s} z_{in} \cdot c_{in}, \\
 & C_4: a_{(i+1)n} = (1 - z_{in})(a_{in} + 1), a_{in} \in \mathbb{N}, \\
 & C_5: 0 \leq a_{in} \leq \lambda, \\
 & C_6: \psi_{in} \in \{0, 1\}, i = 1, 2, \dots, N_s, \\
 & n = 1, 2, \dots, N,
 \end{aligned} \tag{15}$$

where $P1$ is to maximize the ratio of system offered capacity against the request traffic for each cell. $\mathbf{Z} = \{z_{in}\}_{N_s \times N}$ is the optimization variable, which means the schedule result of BH at time slot i . C_1 defines the optimization variable. $z_{in} = 1$ indicates that the cell n is selected. The constraint C_2 restricts the maximum number of beams that the satellite can generate at every time slot. C_3 means total capacity that the system can offered in a period. Inspired by the idea of age of information, we apply the unserved times a_{in} in constraint C_4 , where a_{in} is a non-negative integer. The value of unserved times for cell n at time slot $i + 1$ will be increased by 1 if this cell is not served at time slot $i + 1$. When a cell is selected to be served, its unserved time is reseted to zero. In addition, the unserved times can not exceed the threshold to ensure the fairness of cells, which is guaranteed by constraint C_5 . C_6 denotes the interference pattern of SDBH based satellite, $\psi_{in} = 0$ means the cell should not be illuminated at time slot i .

IV. OPTIMIZATION ALGORITHM OF BEAM HOPPING SCHEDULING

A. Problem Relaxation for TDBH Scheduling

The objective function of this optimization problem is related to two optimization variables $\mathbf{G}$ and $\mathbf{A}$. The beam scheduling result $\mathbf{G}$ is a binary matrix with a large solution space, which makes it difficult to be solved. In this subsection, the original TDBH optimization problem is decomposed into two sub-problems P_a and P_b . In P_a , the minimum value δ is maximized to guarantee the fairness of service. Then time slot allocation will be addressed in P_b . P_a and P_b are expressed as follows.

$$\begin{aligned}
 P_a: \quad & \max_{\mathbf{G}, \mathbf{A}} \delta \\
 \text{s.t.} \quad & C_1: g_{in} \in \{0, 1\}, i = 1, 2, \dots, N_{slot}, \\
 & n = 1, 2, \dots, N, \\
 & C_2: \sum_{n=1}^N g_{in} \leq K, i = 1, 2, \dots, N_{slot}, \\
 & C_3: 0 \leq W_t \leq \tau, \\
 & C_4: \delta \leq A_n \leq 1, \delta \in (0, 1), \\
 & C_5: \phi_{in} \in \{0, 1\}, i = 1, 2, \dots, N_{slot}, \\
 & n = 1, 2, \dots, N.
 \end{aligned} \tag{16}$$

$$\begin{aligned}
 P_b: \quad & \max_{\mathbf{G}, \mathbf{A}} \mathbf{C}_e \cdot \mathbf{T}_e \\
 \text{s.t.} \quad & C_1: g_{in} \in \{0, 1\}, i = 1, 2, \dots, N_{slot}, \\
 & n = 1, 2, \dots, N, \\
 & C_2: \sum_{n=1}^N g_{in} \leq K, i = 1, 2, \dots, N_{slot}, \\
 & C_3: 0 \leq W_t \leq \tau, \\
 & C_4: \rho \delta \leq A_n \leq 1, \rho \in (0, 1), \\
 & C_5: \phi_{in} \in \{0, 1\}, i = 1, 2, \dots, \\
 & N_{slot}, n = 1, 2, \dots, N.
 \end{aligned} \tag{17}$$

where ρ is defined as a system fairness coefficient.

Then, to compress the large solution space brought by the discrete beam scheduling result, the optimization variable $\mathbf{G}$

is relaxed from integer constraint to continuous real number constraint, which can be expressed as

$$\tilde{\mathbf{G}}^{1 \times N} = \left\{ \tilde{G}_n \right\}_1^N, \tilde{G}_n \in (0, 1], \quad (18)$$

where $\tilde{G}_n$ means the ratio of the service time of cell n to a whole beam hopping scheduling period. In addition, since the relaxation solution $\tilde{G}_n$ is a preliminary allocation, therefore, $\mathbf{W}_t$ in (11) is rewritten as follows.

$$\mathbf{W}_t = \frac{1}{\tilde{\mathbf{G}} \circ \mathbf{C} - \mathbf{A} \circ \mathbf{T}}. \quad (19)$$

Then P_a is renewed as follows.

$$\begin{aligned} \tilde{P}_a : \quad & \max_{\tilde{\mathbf{G}}, \mathbf{A}} \quad \delta \\ \text{s.t.} \quad & C_1 : \|\tilde{\mathbf{G}}\|_1 \leq K, \tilde{G}_i \in (0, 1], \\ & C_2 : 0 \leq W_t \leq \tau, \\ & C_3 : \delta \leq A_n \leq 1, \delta \in (0, 1). \end{aligned} \quad (20)$$

After solving $\tilde{P}_a$, we get the value of δ . Then $\tilde{G}$ can be obtained in the renewed problem P_b , which is expressed as

$$\begin{aligned} \tilde{P}_b : \quad & \max_{\tilde{\mathbf{G}}, \mathbf{A}} \quad \mathbf{C}_e \cdot \mathbf{T}_e \\ \text{s.t.} \quad & C_1 : \|\tilde{\mathbf{G}}\|_1 \leq K, \tilde{G}_i \in (0, 1], \\ & C_2 : 0 \leq W_t \leq \tau, \\ & C_3 : \rho\delta \leq A_n \leq 1, \rho \in (0, 1). \end{aligned} \quad (21)$$

Note that both $\tilde{P}_a$ and $\tilde{P}_b$ can be effectively executed by linprog toolbox.

After solving the above problem, we get the ratio of the service time of cell n to an entire beam hopping scheduling period. However, the detailed time slot allocation result under the interference pattern constraint has not been settled yet. Therefore, a time slot allocation strategy is devised to obtain the final TDBH pattern, as detailed in Algorithm 1. Based on the discrete scheduling result $\mathbf{G}$, another optimization variable $\mathbf{A}$ can be obtained by solving the original optimization problem P .

B. Genetic Algorithm for SDBH Scheduling

As described before, the goal of SDBH scheduling is to satisfy the traffic demands of each cell as much as possible. The satisfaction rate is maximized while considering constraints such as unserved times and active beams. In this section, we introduce the optimization algorithm for beam hopping scheduling based on genetic algorithm. First, the definition of main components of the proposed GA algorithm are described.

1) *Representation and Selection*: By considering the layout of two-dimensional hexagonal cells, Fig. 3 illustrates an example of representation of a BH pattern in a given time slot. Each individual is a binary vector representing the activation status (0 or 1) of each cell in a given time slot. The length of the vector corresponds to the number of cells. The number of '1' in each individual is restricted to a maximum of k , due to the constraint on the number of beams that can be active simultaneously. The selection is conducted in proportion to the current population, and an individual is selected under a random strategy.

Algorithm 1 Time Slot Allocation Strategy of Throughput-Driven Beam Hopping Scheduling

Input:

Beam service time, $\bar{\mathbf{G}}$; Number of cells, N ; Number of beams K ;
Number of slots in the scheduling period, T_{num} ;
Lists of cells for restricted and unrestricted services, L_a, L_b ;
List of restricted time slots for cell n , B_n .

Output:

Slot allocation matrix $\mathbf{M} = \{\mathbf{m}_t\}^{T_{num}} = \{m_{n,t}\}^{N \times T_{num}}$, where $\mathbf{m}_t$ denotes the allocation result in time slot t .

Slot allocation:

```

1: Number of time slots to be allocated to cell  $n$ ,  $\mathbf{G}^{slot} = \lfloor \bar{\mathbf{G}} T_{num} \rfloor = \{g_n^{slot}\}^{1 \times N}$ 
2: Initialize the slot allocation matrix  $\mathbf{M}$ 
   Slot allocation for restricted service:
3:   For  $n$  in  $L_a$ :
4:      $t = 1$ 
5:     While  $|\mathbf{m}_t| < K \ \& \ t \leq T_{num} \ \& \ g_n^{slot} > 0 \ \& \ t \notin B_n$ :
6:        $m_{n,t} = 1$ 
7:        $g_n^{slot} = g_n^{slot} - 1$ 
8:        $t = t + 1$ 
9:     End
10:    If  $g_n^{slot} = 0$ , remove  $n$  form  $L_a$ .
11:  End for
   Slot allocation for unrestricted service:
12:  For  $t$  in  $1 : T_{num}$ :
13:    sort  $L_b$  in descending order of the slots number in the  $\mathbf{G}^{slot}$ 
14:    For  $n$  in  $L_b$ :
15:      While  $|\mathbf{m}_t| < K \ \& \ g_n^{slot} > 0$ :
16:         $m_{n,t} = 1$ 
17:         $g_n^{slot} = g_n^{slot} - 1$ 
18:      End
19:      If  $g_n^{slot} = 0$ , remove  $n$  form  $L_b$ .
20:    End for
21:  End for
22: End

```

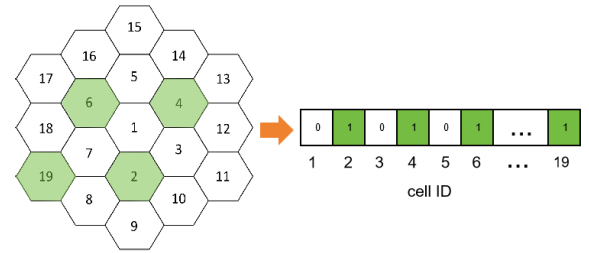


Fig. 3. An example of representation in GA.

2) *Crossover and Mutation*: Pairs of individuals are chosen randomly from the selected subset to undergo single-point crossover. To increase the diversity of the population and promote the algorithm to explore solution spaces, the individual is required to mate according to a crossover probability. As for the mutation operation, this involves randomly flipping the value of bit within the individual with a certain probability.

In this way, genetic diversity is guaranteed and the algorithm is prevented from falling into the local optimal solution. Note that the number of active beams in any mutated individual does not exceed k . Meanwhile, the obtained interference pattern constraint should be satisfied.

3) *Fitness Calculation*: We define the fitness function to evaluate each individual. Considering the optimization goal and unserved times, the fitness function is designed as follows.

If the unserved time is below a certain threshold, the fitness is calculated as the sum of the individual cell satisfaction rates; If the unserved time is above a certain threshold, the fitness incorporates a penalty or adjustment based on the unserved times.

The operation of the GA method which contains the decision-making process and updating process in SDBH scheduling is described in Algorithm 2 in details.

Algorithm 2 The GA Based Satisfaction-rate-driven Beam Hopping Scheduling

Input:

Initialize algorithm parameters including population size p_s , probability of crossover and mutation rate. Set the beam hopping scenario parameters including the number of time slots and cells, the threshold value and traffic demands. Initialize the individuals by creating a population P .

Output: The SDBH time slot scheduling result.

```

1: For time slot  $t = 1$  to  $N_s$  do:
2:   Calculate the channel capacity.
3:   Initialize population with constrains.
4:   For generation  $m = 1$  to  $max\_generation$  do:
5:     Perform selection, crossover, and mutation to creat offspring.
6:     Update population with offspring.
7:     For each individual in population do :
8:       Calculate fitness
9:       Update the best solution if fitness is improved.
10:    End for
11:  End for
12:  Update the unserved times and satisfaction rates.
13: End for

```

V. EVALUATION

A. Simulation Parameters

In this section, simulation experiments are conducted to evaluate the problem relaxation-based algorithm and GA-based BH scheduling algorithm. Simulations of the relaxation based algorithm for TDBH scheduling are performed on Matlab platform and the parameters are summarized in Table I. We assume that the TDBH simulation scenario is a GEO multi-beam satellite. Some parameters refer to the simulation assumptions in TR38.821 [26], such as 3dB antenna beam width, transmitting gain, satellite antenna aperture of Ka-band. The receiving gain is set to 0 dBi. Assume that the GEO satellite covers 19 cells and the maximum number of beams equals 4. There are 33 time slots in the whole scheduling period. The uniform distribution parameter of the TDBH scheduling result is set to $U(500, 1500)$.

TABLE I
SIMULATION PARAMETERS FOR TDBH

PARAMETERS	VALUES
COMMUNICATION PARAMETERS	
Bandwith of beams B	500 MHz
Satellite orbital altitude L	36000 km
3dB antenna beam width θ_{3dB}	0.17°
Satellite antenna aperture	2 m
Transmitting power of each beam P_t	20 dBW
Carrier frequency f_c	20 GHz
Equivalent noise temperature T_{noise}	300K
Transmitting gain G_t	50 dBi
Receiving gain G_r	0 dBi
Number of virtual cells n	19
Demands distribution	$U(500, 1500)$
Period of BH scheduling	33
ALGORITHM PARAMETERS	
Fairness coefficient ρ	0.8

TABLE II
SIMULATION PARAMETERS FOR SDBH

PARAMETERS	VALUES
COMMUNICATION PARAMETERS	
Bandwith of beams B	500 MHz
Satellite orbital altitude L	600 km
3dB antenna beam width θ_{3dB}	4.6°
Satellite antenna aperture	0.2 m
Transmitting power of each beam P_t	20 dBW
Carrier frequency f_c	20 GHz
Equivalent noise temperature T_{noise}	300K
Transmitting gain G_t	50 dBi
Receiving gain G_r	0 dBi
Number of virtual cells n	19
Demands distribution	$U(100, 1000)$
Time slot duration	2 ms
Period of BH scheduling	21
ALGORITHM PARAMETERS	
Mutation rate	0.02
Crossover probability	0.7
Population size	100
Max generation	50

The GA-based algorithm for SDBH scheduling simulations is performed on Python 3.8 platform and the parameters are summarized in Table II. The SDBH simulation scenario is a typical Ka-band LEO multi-beam satellite, where the bandwidth of each beam is 500 MHz, the number of service virtual cells is 19, and the maximum number of satellite beams within each slot is 4. The coverage of the beam is determined by the antenna gain 3dB width which refers to the 3GPP TR 38.821 [31], where $\theta_{3dB} = 4.6^\circ$. To facilitate the following analysis, the transmitting power is set to be equally shared among cells. It should be noted that the traffic demands are set to be uniformly distributed. The downlink Shadowed-Rician fading channel can be infrequent light shadowing, frequent heavy shadowing and average shadowing [30]. There are three parameters to describe Shadowed-Rician fading channel, average power of the NLOS component b_0 , the Nakagami-m fading parameter m , the average power of the LOS component Ω . The Shadowed-Rician parameters corresponding to the fading severity are shown in in Table III. Fig. 4 and Fig. 5 show the interference pattern of the GEO satellite

TABLE III
THE SHADOWED-RICIAN PARAMETERS

Shadowed-Rician channels	b_0	m	Ω
Light shadowing	0.158	19.4	1.29
Heavy shadowing	0.063	0.739	8.97×10^{-4}
Average shadowing	0.251	5.21	0.278

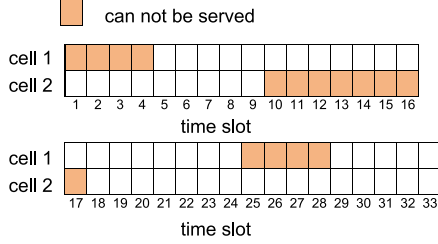


Fig. 4. GEO satellite interference pattern of TDBH scheduling.

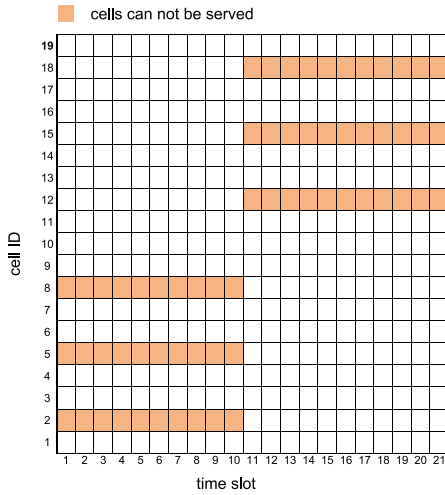


Fig. 5. LEO satellite interference pattern of SDBH scheduling.

and the given interference pattern for the LEO satellite, respectively.

B. Problem Relaxation Based Algorithm Performance

It can be seen in Fig. 4 that only the illumination of cell 1 and cell 2 of the TDBH based GEO satellite are constrained by the interference. Based on the GEO satellite interference pattern, we first get the preliminary allocation result through relaxation, which is shown in Fig. 6. Then we obtain the discrete time slot allocation result for the GEO satellite through the time slot allocation strategy, that is, the GEO BH pattern, which is shown in Fig. 7. Fig. 8 illustrates the adaptive rate adjustment factor after solving the original problem based on discrete time slot allocation result. It can be seen that there is a minimum value of the adaptive rate adjustment factor, 0.32594. Therefore, regardless of the gap between cell channel conditions and traffic demands, the adaptive rate adjustment factor of the cells will not be lower than the minimum value, ensuring service fairness. Fig. 9 shows the GEO system throughput result of TDBH scheduling.

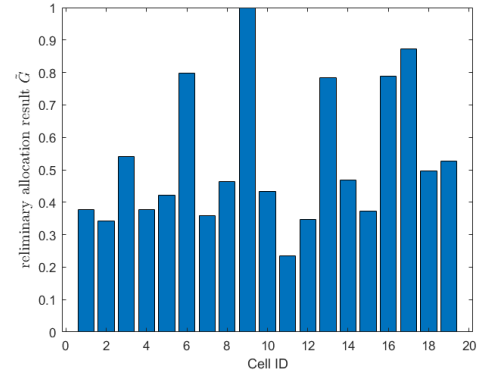


Fig. 6. Preliminary allocation result of TDBH scheduling.

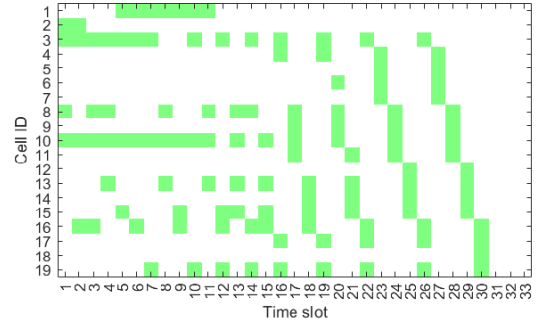


Fig. 7. Discrete time slots allocation result of TDBH scheduling.

As mentioned before, the traffic demands follow a uniform distribution. Here, we define an intensity coefficient of the traffic demands as $T_n \sim M \cdot U(500, 1500)$. Then its mean value is expressed as $E[T_n] = M \cdot 1000 \text{ Mbit/s}$. Besides, three shadowing conditions of the satellite link are considered, which are infrequent light shadowing, frequent heavy shadowing, and average shadowing [30]. In Fig. 10, we investigate the impact of traffic load intensity and different channel conditions as well as scheduling methods on the system throughput of the GEO satellite. We can observe that the traffic load intensity has little impact on the system throughput for heavy shadow fading channel. Both the proposed scheduling method and random scheduling have poor performance for the heavy shadow fading channel, though the system throughput of the proposed scheduling method outperforms that of the random scheduling. As for the light shadow fading channel and average shadow fading channel, when M is below a specific value, the system throughput increases with M . However, the system throughput will get saturated when $M > 3$, because the system throughput will depend on the channel capacity and the adaptive rate adjustment factor then, when the traffic load exceeds the channel capacity.

In addition, our method schedules beams with poor channel capacity due to shadowing effects, which is the key to improving system throughput under the same traffic load intensity value. For the heavy shadowing channel, our proposed method allocates the beams with poor channel conditions to cells with relatively low service traffic demands, avoiding significant adaptive rate adjustment and thus improving the system throughput. In contrast, random scheduling may assign

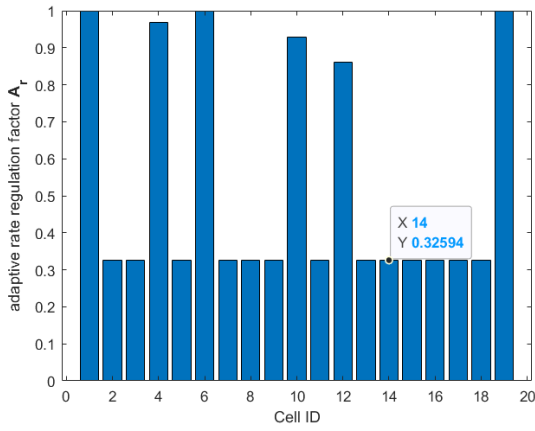


Fig. 8. Adaptive rate adjustment factor result.

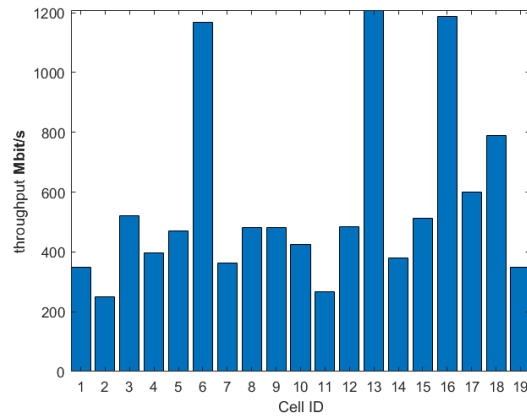


Fig. 9. GEO satellite system throughput of TDBH.

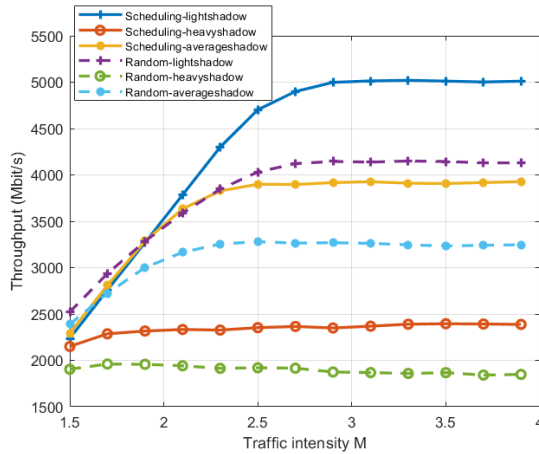


Fig. 10. TDBH scheduling system throughput performance of GEO.

beams to high-traffic demand cells with low channel capacity. Therefore it will have a small value of the rate adjustment factor, thus reducing the system throughput.

C. Genetic Algorithm Performance

The SDBH scheduling result of the LEO satellite is based on the interference pattern shown in Fig. 5, which aims to guarantee the service quality of GEO satellite and terrestrial

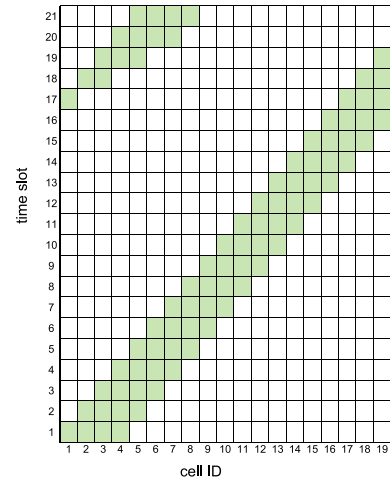


Fig. 11. A RRBH pattern.

network. We define the average traffic satisfaction rate for each cell $\alpha_n = \min(R_n, D_n)/D_n$ and the average total satisfaction rate of a satellite $\beta = \sum_{n=1}^{N_i} \alpha_n/N_i$ to evaluate the performance of different beam hopping algorithms. To verify the impact of the genetic algorithm on the performance of LEO satellite beam hopping scheduling, we first compare the proposed method with four different BH schemes as follows.

- The Random Beam Hopping (RBH): The RBH randomly selects 4 cells at each time slot from the 19 cells.
- The Round Robin Beam Hopping (RRBH): The RRBH method means a satellite serves the covered cells in a specific order. In this simulation, the RRBH takes a beam hopping pattern where the satellite covers 19 virtual cells and generates 4 beams at each time slot. Fig. 10 shows a RRBH pattern, which presents the cells that are illuminated by beams. It should be noted that some cells could not be served due to the interference pattern.
- The Greedy Demand-Based Beam Hopping (GDBH): According to [32], the GDBH selects top 4 cells with the largest demand at each time slot to serve according to the rank of obtained traffic demands.
- The Genetic Algorithm Beam Hopping (GABH): The satellite makes the beam plan with the genetic algorithm adopted in [33], in which fairness constraint is not considered.

The average normalized traffic satisfaction rate of 19 cells is shown in Fig. 12, and the average total satisfaction rates are summarized in Table III. It can be seen from Fig. 12 that the proposed GA-based method cannot achieve the maximum value for every cell since it considers the fairness, however, it achieves the highest average total satisfaction rate. From Table III, compared with RRBH, RBH, GABH and GDBH, the average traffic satisfaction rate of the proposed GA method is improved by 41.36%, 25.23%, 16.96% and 46.90%, respectively. The RRBH, RBH and GDBH scheduling strategies are relatively simple, but they perform worse than the GABH and the proposed method.

Furthermore, the minimum average rate of the proposed GA method is 23.58%, which is higher than the minimum value of

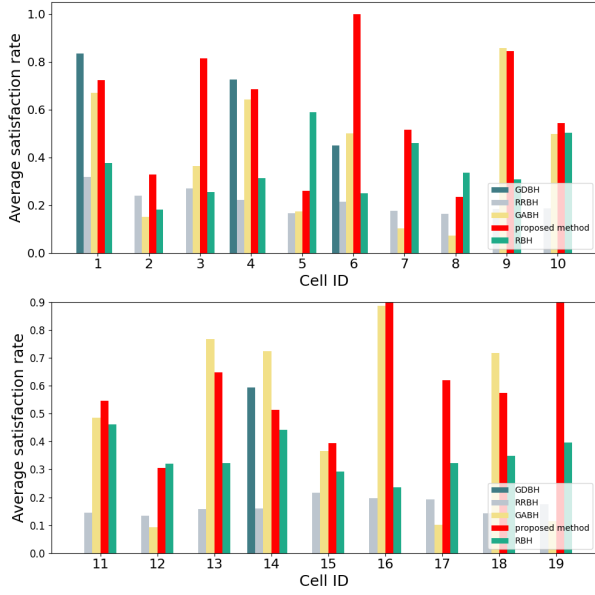


Fig. 12. Average traffic satisfaction rate of 19 cells among different algorithms.

TABLE IV
AVERAGE TOTAL TRAFFIC SATISFACTION RATE RESULTS

ALGORITHM	AVERAGE TOTAL SATISFACTION RATE	MINIMUM AVERAGE SATISFACTION RATE
RRBH	19.24%	13.34%
RBH	35.37%	18.23%
GABH	43.64%	7.27%
GDBH	13.70%	0
Proposed GABH method	60.60%	23.58%

GABH, 7.27%, while the other three minimum average rates of RRBH, RBH, and GDBH are 13.34%, 18.23%, 0, respectively. Since the proposed GA method considers service fairness, it has a better performance on the minimum average rate than GABH. For high-traffic demand cells 1, 4, 6, 14 which GDBH focuses to serve, they also have been effectively satisfied in our method. Since RBH and RRBH completely ignore the traffic demands, the satisfaction rate of high-traffic demand cells is relatively low, which means the differentiated traffic demand is not satisfied effectively. GDBH only selects cells with high demand at every slot and the demand of each cell is unchanged during the period, resulting in a 0 satisfaction rate in other 15 cells. In addition, GABH only optimizes each cell independently, while ignoring the fairness of cells. As a result, it performs worse with respect to average total satisfaction rate than the proposed method.

We further evaluate the proposed method under different demand distributions with the same mean value of 550 Mbps but different variances. Then different channel fading conditions are considered, aiming to better illustrate the effectiveness. The numerical results of three uniform distributions with different variances are shown in Fig. 13. It is observed that the proposed GA method can effectively adapt to demand

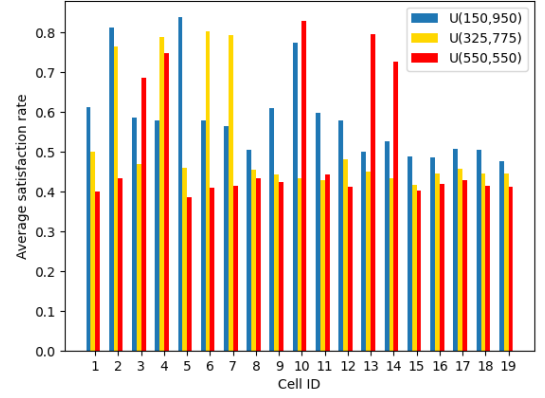


Fig. 13. Average traffic satisfaction rate of 19 cells among different distributions.

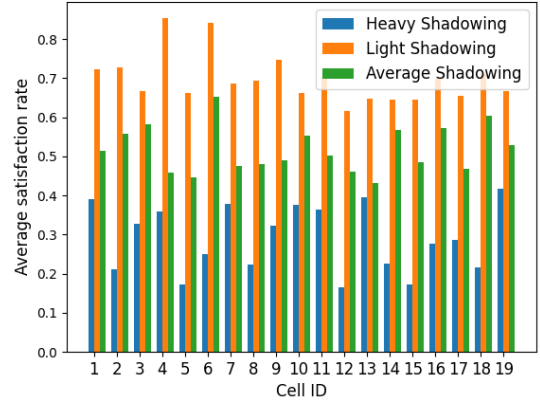


Fig. 14. Average traffic satisfaction rate of 19 cells among different channels.

distributions with different variances, where the total average satisfaction rate $\beta_{U(150,950)} = 62.63\%$, $\beta_{U(325,775)} = 52.70\%$, $\beta_{U(550,550)} = 52.83\%$, thus variances have limited impact on the proposed method, and the proposed GA scheduling method achieves a good performance while meeting the non-uniform traffic requirements. The average satisfaction rates of different fading channels are shown in Fig. 14, where we consider the light shadow, the average shadow and the heavy shadow fading under the same traffic demand distribution $U(100,1000)$. It is obvious that the average satisfaction rates of the heavy shadow fading channel are much lower compared with the light shadowing due to the limited channel capacity, therefore the proposed method is suited to scenarios with different kinds of traffic demands and fading channel conditions.

VI. CONCLUSION

In this paper, we have studied the scheduling optimization problems of satellite beam hopping for interference avoidance in the integrated satellite-terrestrial wireless networks. At first, the two optimization problems are formulated, respectively. The TDBH scheduling problem aims to maximize the satellite throughput and we consider the traffic demands can be regulated in the TDBH scenario. While satellites which employ SDBH focus on serving cells whose traffic demands can not be altered and the optimization goal is to maximize the system satisfaction rate. Then we decompose the original

TDBH scheduling problem into two sub-problems by adopting relaxation, and genetic algorithm is introduced to handle the SDBH scheduling problem. The simulations involve a TDBH based GEO satellite and a SDBH based LEO satellite. It is worth noting that the SDBH scheduling is based on the TDBH scheduling because GEO satellites take priority over LEO satellites in terms of the BH scheduling of our system. The simulation of TDBH scheduling analyzes the impact of three channel conditions and traffic load intensity on the performance of satellite system throughput. As for SDBH scheduling, the simulation has shown that the proposed GA method achieves the highest average total satisfaction rate and minimum average satisfaction rate, thus the cells with low traffic demands can be well served. In addition, the GA method is suited to scenarios with different kinds of traffic demands and fading channel conditions.

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